



GAME DESIGN DOCUMENT

>EXIT<

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Table of Contents

1. Game Overview.....	4
1.1. Game Concept.....	4
1.2. Genre.....	4
1.3. Target Audience.....	4
1.4. Game Flow Summary.....	5
1.5. Look and Feel.....	5
2. Gameplay and Mechanics.....	5
2.1. Gameplay.....	5
2.1.1. Game Progression.....	6
2.1.2. Mission/Challenge Structure.....	6
2.1.3. Puzzle Structure.....	6
2.1.4. Objectives.....	6
2.1.5. Play Flow.....	6
2.2. Mechanics	6
2.2.1. Physics.....	7
2.2.2. Movement in the game.....	7
2.2.3. Objects.....	7
2.2.4. Actions.....	7
2.2.5. Combat.....	7
2.2.6. Economy.....	7
2.2.7. Screen Flow.....	7
2.3. Game Options.....	8
2.4. Replaying and Saving.....	9
2.5. Cheats and Easter Eggs.....	9
3. Story, Setting and Character.....	9
3.1.1. Plot Elements.....	9
3.1.2. Game Progression.....	9
3.1.3. Cut Scenes.....	10
3.2. Game World.....	10
3.2.1. General look and feel of the world.....	10
3.2.2. Areas - incomplete.....	10
4. Levels.....	11
4.1. Levels.....	11
4.2. Training Level.....	12
5. Interface.....	12
5.1. Visual System.....	12
5.2. Control System.....	13

5.3. Audio, music, sound effects.....	13
5.4. Help System.....	13
6. Artificial Intelligence [Requires Research].....	13
6.1. Opponent and Enemy AI.....	13
6.2. Non-combat and Friendly Characters.....	13
6.3. Support AI.....	13
7. Technical.....	14
7.1. Target Hardware.....	14
7.2. Development Hardware and Software.....	14
7.3. Network requirements.....	14
8. Game Art.....	14

1. Game Overview

A magician wakes up in a castle and needs to progress through the castle to escape its confines. As the magician progresses through the castle, he encounters ghosts that attempt to prevent him from progressing through. To progress through the rooms of the castle, he finds keys that allow him to move from room to room.

Concept Ideas:

- Third person
- Sphere monsters - Ghosts
- Character can run from ghosts
- Powerups
- Ghosts have health
- //Boss fights? Required to kill in order to a key [TBD]
- Magician
- Wand shoots sparks and kills ghosts
- Rooms
- Collect keys to open the door for the room
- Character wakes up in a castle
- Character picks up items to learn
- At the end of the game, he picks up a letter that explains everything

1.1. Game Concept

Players take on the role of a skilled magician who wakes up in the castle with no memory of how they got there, they must navigate through the rooms to find keys and to escape.

1.2. Genre

The genres of the game will be fantasy, mystery and discovery, with the goal being to escape.

1.3. Target Audience

The target audience is teens and above (13+).



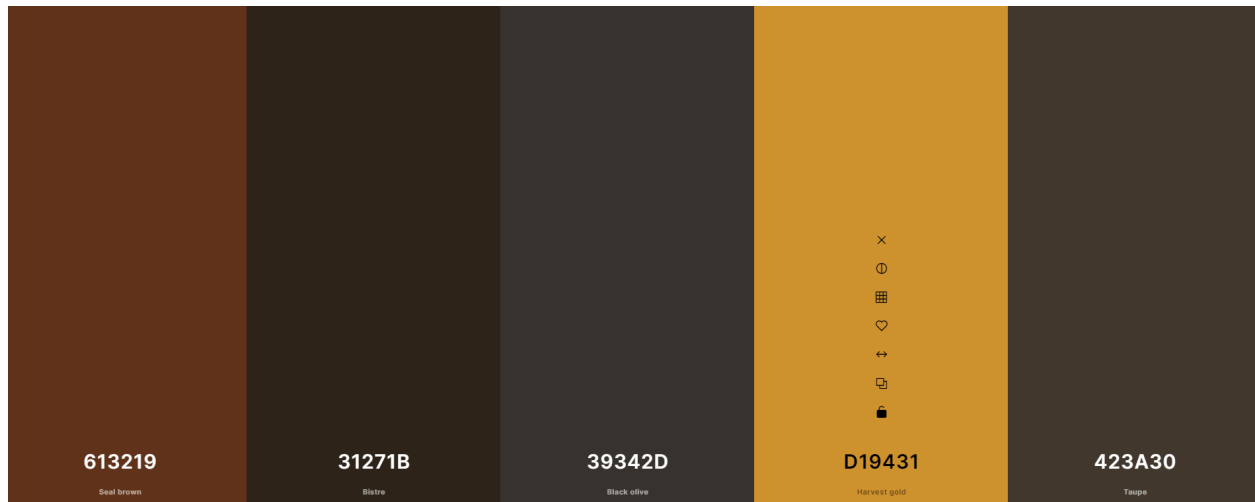
1.4. Game Flow Summary

The game starts at the Menu Page, where the user/player needs to click on 'PlayGame' to start the game. When the player presses 'PlayGame', there is a loading screen before the player starts in the first level. When the player starts in the first level, there is a popup that shows what the controls of the game are. The player moves

1.5. Look and Feel

The game's look and feel are designed to immerse players in a fantasy world with magic, mystery, and danger.

Color Palette:



2. Gameplay and Mechanics

2.1. Gameplay

The character moves through the rooms while being targeted by ghosts who try to kill him. The character fights the ghosts back by shooting sparks from his wand. Occasionally, the character can pick up power ups that allow him to gain temporary powers, such as increasing his range of vision. The player finds keys to open the door to the next level. The player finds items to permanently upgrade his powers that allow him to more easily progress through the level.

2.1.1. Game Progression

The player finds keys to open the Door to proceed with the next level.

2.1.2. Mission/Challenge Structure

Each level presents a new challenge / harder set of obstacles making it harder to progress. The magician must navigate through the castle, finding keys to unlock the doors and go to the next area. The ghosts are the main obstacles all with different looks and abilities, the magician must outsmart the mobs.

2.1.3. Puzzle Structure

We have no plans of adding a puzzle to the game.

2.1.4. Objectives

Get through each level unscathed, gather every key to unlock the doors and escape, uncover the hidden truth, and ultimately conquer the game.

2.1.5. Play Flow

The player starts at the main menu page and clicks play to start:

- A loading screen may appear before the first level;
- A small popup shows the game controls when the first beginning level starts;
- The player must now move through the rooms with the goal of defeating the ghosts, finding the keys to progress.

2.2. Mechanics

What are the rules to the game, both implicit and explicit. This is the model of the universe that the game works under. Think of it as a simulation of a world, how do all the pieces interact? This actually can be a very large section.

- The player is allowed to pick up power ups, which allows them to acquire a certain power temporarily. For example, one of the power-ups allow the magician to gain a bigger range of vision.
- The player can also pick up items that allow them to receive permanent upgrades for their character. [Upgrades TBD]
- The player must find keys in order to open the door to the next rooms/levels.

2.2.1. Physics

Other than the fantasy aspects in our game, the game's physics are based on real-world physics principles. Objects behave according to standard physics rules, including gravity, collision detection, and object interactions. This makes sure players can expect non-fantasy objects to behave as they would in the real world.

2.2.2. Movement in the game

The player can either walk or dash in the game. Dashing allows the player to progress faster through the levels.

2.2.3. Objects

You can press E to pick up the powerups and the items that give you permanent upgrades

2.2.4. Actions

- **Player Movement:** (WASD) AND/OR (UP-RIGHT-DOWN-LEFT) ARROWS
 - **Opening Doors:** E || F (TBD)
- Possible Concepts:**
- // Reload Ward Stick after 20 shots?
 - // Jump over obstacles like table or chair (space)
 - // Dash (x)

2.2.5. Combat

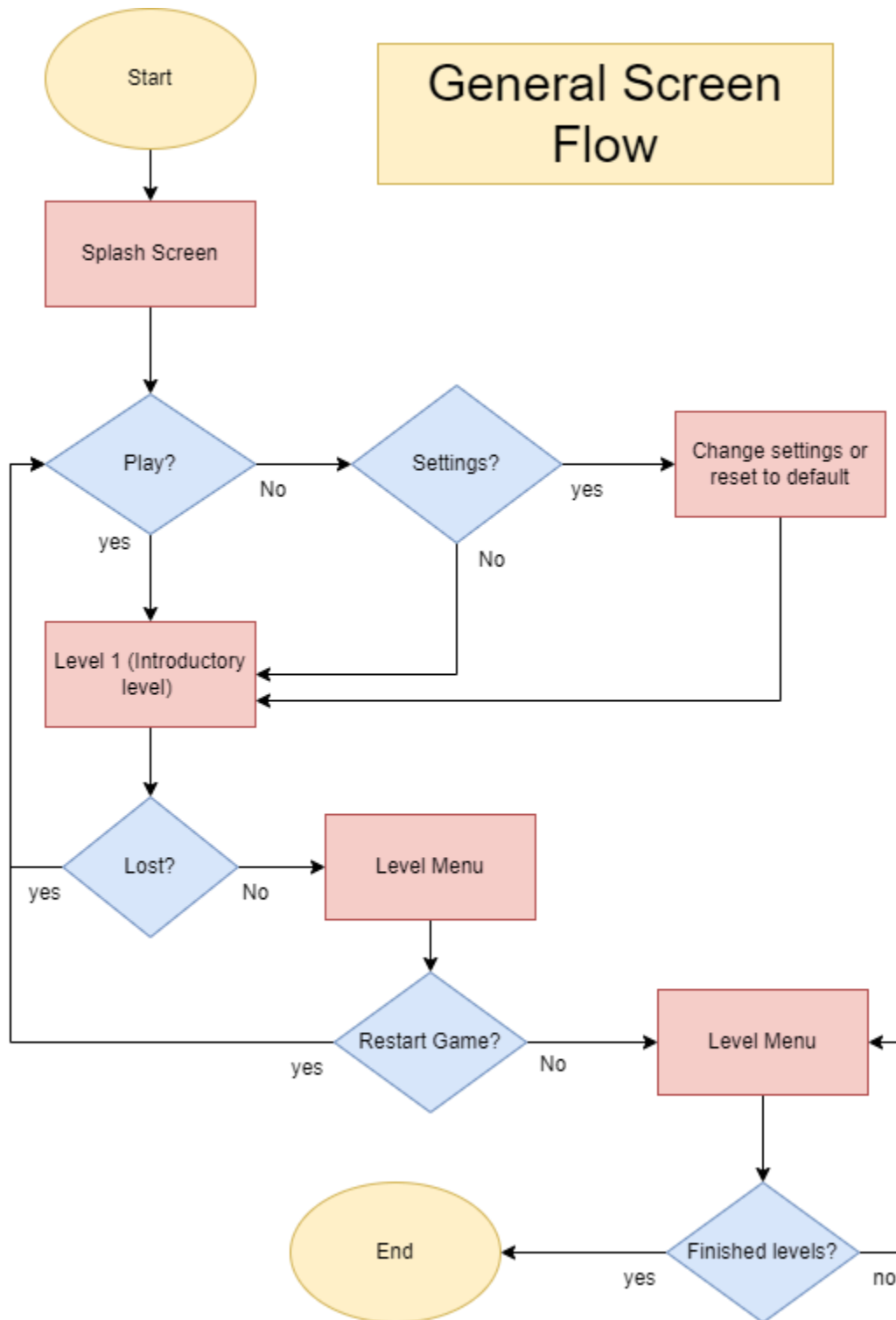
Combat involves the magician using his wand at the ghosts. Each ghost has a health bar, and the magician must reduce it to zero to defeat them. [The combat controls are TBD.]

2.2.6. Economy

There is no currency in the game, however, there will be powerups.

2.2.7. Screen Flow

Screen flow is represented by a flowchart, which can be found here: [Flow Chart](#).



Note: Each time the magician opens a door and enters a new room, it is considered a new level.

2.3.Game Options

Combat Tactics:

- The player can choose their approach to combat, fight the ghosts or run/avoid contact with ghosts.
- The player can choose to pick up and use power ups to fight ghosts.

Exploration Paths:

- The player can choose to roam around the room searching for the keys first, or fight the ghosts and then search for the keys.

2.4.Replaying and Saving

After a level is beaten, a checkpoint at the start of each level is automatically created. That way, if the user fails a level, they can go back to it.

2.5.Cheats and Easter Eggs

The character can pick up cool shades!



3. Story, Setting and Character

3.1.Story and Narrative

3.1.1. Plot Elements

- **Introduction:** The magician awakens in a castle, realizing he is trapped.
- **Objective:** He does not recognize his surroundings and realizes he must escape.
- **Obstacles:** The path is blocked by ghosts. Ghosts lurk in various rooms - hindering progress.
- **Key Mechanic:** To move forward, he must find keys hidden within the castle's rooms. With each key found, he inches closer to the exit and to his freedom, defeating ghosts along the way.

3.1.2. Game Progression

- **Starting Point:** The magician begins in a dimly lit chamber.
- **Exploration:** He explores adjacent rooms, encountering weaker ghosts initially.
- **Key Discovery:** Keys are found lying in various places in the room.

- **Unlocking Doors:** Using keys, the magician unlocks doors to access new areas.
- **Increasing Difficulty:** Ghosts get stronger as the levels progress
- **Escape:** Upon defeating the final challenge, the magician escapes the castle, achieving victory.

3.1.3. Cut Scenes

- **Introduction:** The magician waking up in the castle, realizing he's trapped.
- **Key Discovery:** Finding a key(s).
- **Escape:** The magician emerges from the castle's gates as sunlight breaks through the clouds, signaling freedom.

3.2. Game World

The game is situated in a dark castle. This dark castle contains rooms through which the player must progress through.

3.2.1. General look and feel of the world

The game world is a dark castle with stone walls, dim lighting, and mysterious ambiance. There is also some just creepy unsettling but still adventurous music playing in the background. Each room offers a different challenge or puzzle for the player to overcome.

3.2.2. Areas - incomplete

The castle is divided into various areas, each with its own unique characteristics and challenges. Some areas may contain stronger ghosts or require more keys to progress.

Includes the general description and physical characteristics as well as how it relates to the rest of the world (what levels use it, how it connects to other areas)

3.3. Characters

Each character should include the back story, personality, appearance, animations, abilities, relevance to the story and relationship to other characters

- ★ Magician - main character
 - Sphere (color: TBD [based on assets])
 - Magician hat

- Magic Wand - Shoots sparks towards the Ghosts, causing damage to the Ghosts.
- ★ Ghosts - enemies
 - Sphere (color: TBD [based on assets])
 - Glows (optional)
 - Cause damage to the main character upon contact

4. Levels

4.1. Levels

Level 1:

Synopsis: The magician awakens in a dimly lit chamber, realizing he's trapped in the castle. This level serves as the introductory stage, introducing basic mechanics and setting the tone for the game.

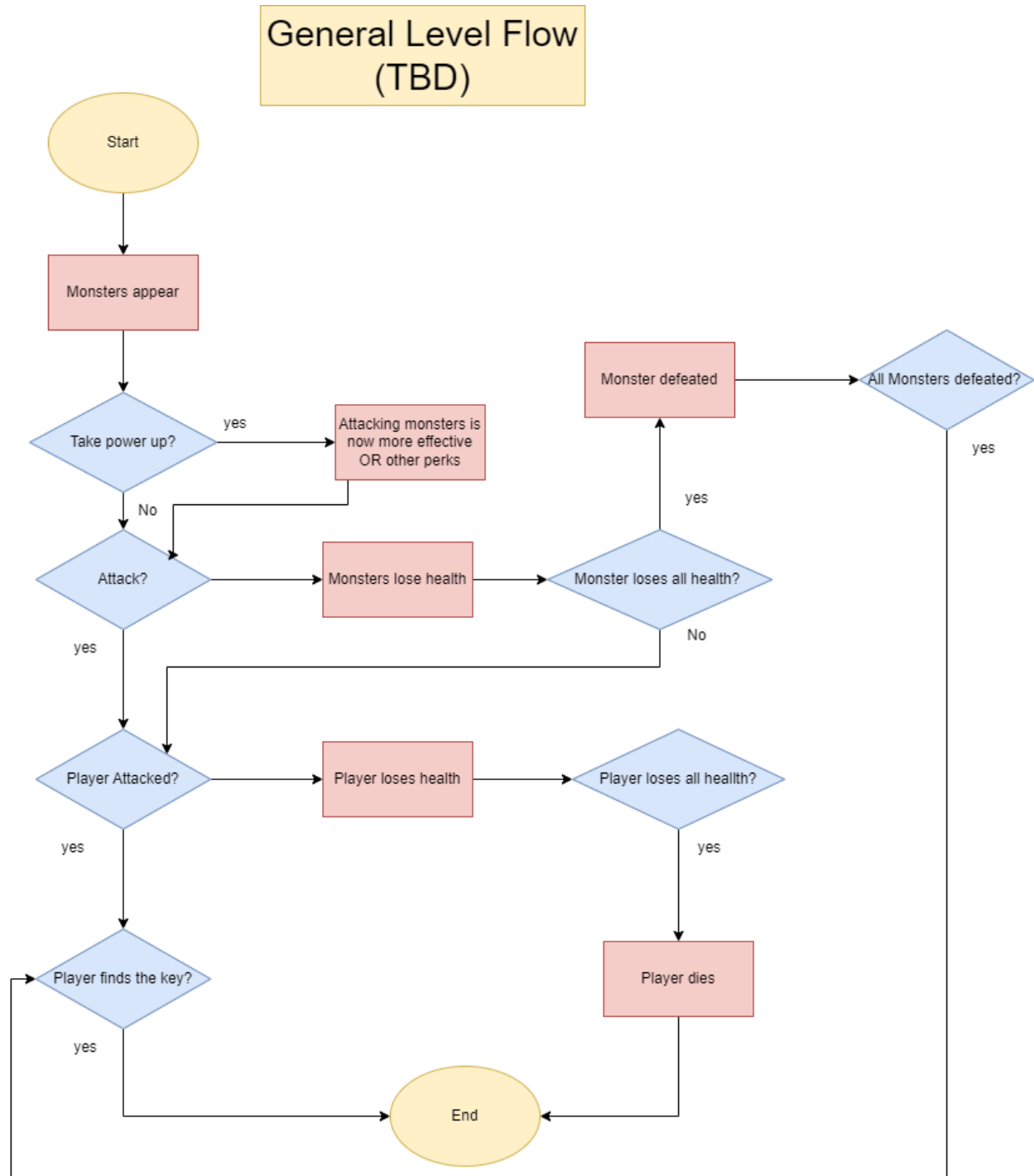
Introductory Material: Upon awakening, a brief cutscene shows the magician waking up and looking at his surroundings.

Objectives:

1. Explore the chamber to familiarize yourself with controls and mechanics.
2. Locate the first key hidden within the room.

Details:

- The chamber is relatively small, with stone walls.
- The magician can interact with objects.
- Encounters with ghosts- serve as introductory combat sequences.
 - Magician can shoot sparks from his magic wand at the ghosts- causing damage to the ghosts.
- The first key is placed somewhere on the floor, the magician must search for it.
- Collecting the key will open the Door, progressing the game -> leading to the next level.



4.2. Training Level

Non-applicable

5. Interface

5.1. Visual System

HUD Content

- There will be a health bar for the player as well as for the ghosts. Power ups that can be picked by the player will allow the player to progress through the level more easily. For example, the player could do more damage to the ghosts, suffer less damage when they are hit, heal, become faster, etc [TBD]
- In order to complete a level, the player must find the key to the next room, which is its objective.

Menus

- The attack and music volumes will be able to be lowered or heightened. [TBD]

Camera Model

- The game will be played in 3rd person perspective from above.

5.2. Control System

- Keyboard commands: 'ASDW' & 'UP, DOWN, LEFT, RIGHT' Arrows
- Mouse commands: Clicks

5.3. Audio, music, sound effects

Music:

- Background music

Sound effects:

- Shooting (from wand)
- Walking (magician)
- Getting hit (Magician and Ghosts)
- Ghost (Sighting)

TBD:

- Interactions with doors (opening/closing)
- Moving from one level to another
- Etc.

5.4. Help System

The only help system in the game is the popup at the start of the game which tells the player the controls used to play the game and the main task of the game.

6. Artificial Intelligence [Requires Research]

6.1. Opponent and Enemy AI

The active opponent that plays against the game player and therefore requires strategic decision-making. The enemy AI in the game are the ghosts that roam the castle. The ghosts chase after the player once the player is in the vicinity. (attacks?)

6.2. Non-combat and Friendly Characters

There are no non-combat/friendly characters in the game.

6.3. Support AI

Support AI will possibly include pathfinding for ghosts -> Player in the game world.

7. Technical

7.1. Target Hardware

Compatible with PC equivalent or superior chip to Intel i5 and can run at least 30 frames per second. The game should be able to run on a mobile device if we theoretically added mobile compatibility.

7.2. Development Hardware and Software

We will use Unity as the game engine, Blender to create 3D assets and the Unity 3D assets store to find free 3D assets to use in our game.

- Unity
- Blender,
- Unity3DAssets
- [Sound Effects]
- [Music]

7.3. Network requirements

Non-applicable

8. Game Art

The game art will include assets from the Modular Castle asset pack. This pack provides various castle-themed assets that can be used to create the game's environments. [Link To Asset Package](#)

